FLE TRIBE

WEEK 12 · WEDNESDAY · LESSON GUIDE

Lists, conditions & events

This week: Create a React app with Vite and build UI from components, props and state.

Rendering lists

Use `map` to turn an array of data into an array of elements. Every item needs a `key` that is unique and stable, usually the item's id.

```
function ProductList({ products }) {
  return (
    <ul>
      {products.map((p) => (
        <li key={p.id}>{p.name}</li>
      ))}
    </ul>
  );
}
```

Avoid using the array index as a key when items can be added, removed or reordered: React may mix up which item is which.

Conditional rendering

```
{isLoading && <Spinner />}
{error ? <ErrorMessage text={error} /> : <Results items={items} />}
{items.length === 0 && <p>No products match your search.</p>}
```

Watch out: `{count && …}` renders a **0** when count is 0. Use `{count > 0 && …}` instead.

Handling events

```
<button onClick={() => setOpen(true)}>Open</button>
<input value={query} onChange={(e) => setQuery(e.target.value)} />
<form onSubmit={(e) => { e.preventDefault(); save(); }}>…</form>
```

Pass a function, don't call it: `onClick={save}` or `onClick={() => save(id)}`, never `onClick={save()}`.

Research questions

Answer these in the app, in your own words, to unlock the next lesson.

1. What goes wrong if you use the array index as a key?
2. Build a `ProductCard` list from an array.

Start your research here

- react.dev: https://react.dev/learn/rendering-lists
- react.dev: https://react.dev/learn/responding-to-events